AML/DL

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Hidden Layers

PREDICTIVE SALES ANALYTICS ENGINE

Team Members:

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PROBLEM STATEMENT & MOTIVATION

PROBLEM STATEMENT

- Sales teams rely heavily on intuition and manual judgment
- Difficult to predict whether a deal will succeed or fail
- Large amount of data (emails, notes, interactions) is underutilized

MOTIVATION

- Need for a data-driven system to improve decision-making
- Early prediction can help prioritize high-value opportunities
- Reduce time, effort, and revenue loss from failed deals

WORK PLANNING, TEAM ROLES & GITHUB WORKFLOW

TEAM ROLES (PHASE 1)

Member	Role	Focus Area
Vivekananda	Technical Lead	Repo setup, EDA, feature engineering, model development & evaluation
Manasa	Research Lead	Literature review, dataset selection, theoretical rigor, report & presentation

PROJECT MANAGEMENT & STRUCTURE

- Maintaining GitHub Project Board for task tracking
- Planning to maintain a progress log for future iterations
- Organized and modular folder structure for scalability

GITHUB WORKFLOW

- Project Board → Tasks tracked as Backlog / To-Do / In-Progress / Done
- Issues → Each task documented with owner & deadline
- Branch Naming → name/issue-number-description
- Daily Review → Sync progress + PR reviews

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LITERATURE REVIEW

TRADITIONAL APPROACHES

- Bag-of-Words + Logistic Regression
- Uses simple text features
- Interpretable but ignores context

DEEP LEARNING APPROACHES

- Multimodal models (text + numerical data)
- Capture semantic meaning and patterns
- High complexity and training cost

HYBRID / ENSEMBLE METHODS

- Combine NLP embeddings with ML models
- Best performance in many cases
- Higher complexity and lower interpretability

KEY INSIGHTS FROM LITERATURE

- Text + structured data both are important
- No single model is sufficient alone
- Trade-off between accuracy and interpretability

REFERENCES

- *Perelman & Grinberg (2013)*
- *Shi et al. (2021)*
- *Koval et al. (2024)*

GAPS IN EXISTING APPROACHES

- Lack of effective fusion between text and structured data
- Models are either too simple or too complex
- Limited focus on interpretability

REAL-WORLD CHALLENGES

- Sales data is noisy and unstructured
- Data imbalance (more losses than wins)
- Risk of data leakage in conversation data

KEY PROBLEM

Existing solutions are not fully suitable for real-world sales scenarios

This motivates the need for a robust and practical predictive system.

OBJECTIVE / WHAT WE BUILT

OBJECTIVE

- Build a system to predict sales outcomes (win/loss)
- Use both text data and structured CRM features
- Improve decision-making using data-driven insights

WHAT WE BUILT

- Combined text features (emails, notes) with tabular data
- Applied machine learning models for prediction
- Designed a pipeline from data processing → modeling → evaluation

KEY FOCUS

- Balance between performance and interpretability
- Handle real-world challenges (noise, imbalance, data leakage)

DATASET OVERVIEW

Datasets Used

- Sales Conversations Dataset (text-based interactions)
- CRM Sales Opportunities Dataset (structured pipeline data)

Data Types

- Text Data:
  - Full conversation text
  - Emails / chat interactions
  - Conversation style & flow
- Structured Data:
  - Deal stage, close value
  - Customer & product details
  - Engagement and effectiveness scores

Target Variable

- Binary outcome: Win (1) / Loss (0)

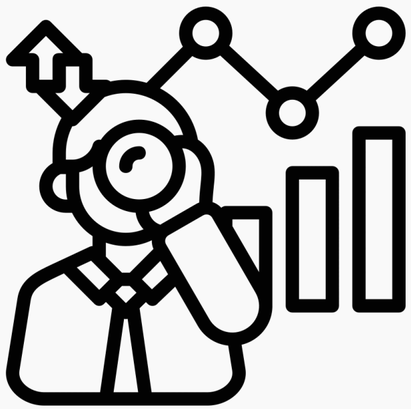
Key Characteristics

- High-dimensional text embeddings (3072 features)
- Imbalanced data (fewer wins than losses)
- Mix of structured + unstructured data
- Temporal and conversational patterns present

EXPLORATORY DATA ANALYSIS (EDA)

- **Target Distribution**
  - Target variable is well balanced (Win ≈ Loss)
  - No need for resampling techniques
- **Feature Relationships**
  - Sales effectiveness (0.86) and customer engagement (0.78) show strong positive correlation with outcome
  - These are key predictive features
  - Conversation length has very weak correlation (-0.04)
- **Text & Behavioral Insights**
  - Higher engagement scores are concentrated around 0.6 – 0.75 range
  - Conversation length mostly lies between 10–14 messages
  - Text length shows slight variation between win/loss (not a strong separator)
- **Data Quality Observations**
  - No major outliers in sales effectiveness (stable distribution)
  - Moderate correlation between conversation length and text length (~0.5)
  - High-dimensional features require careful handling

FEATURE ENGINEERING



CATEGORICAL FEATURES

- One-hot encoding
- Removed redundant categories

TEXT FEATURES

- Text length & word count
- Sentence embeddings (MiniLM – 384D)

ENGINEERED FEATURES

- Engagement per turn
- Engagement × conversation length

TRANSFORMATIONS

- Log scaling for skewed features
- Standardization (scaled features)

DIMENSIONALITY REDUCTION

- PCA → reduced embeddings (384 → 20)
- Retained ~69% variance

NOTEBOOK

BASELINE APPROACH: SUPPORT VECTOR MACHINE (SVM)

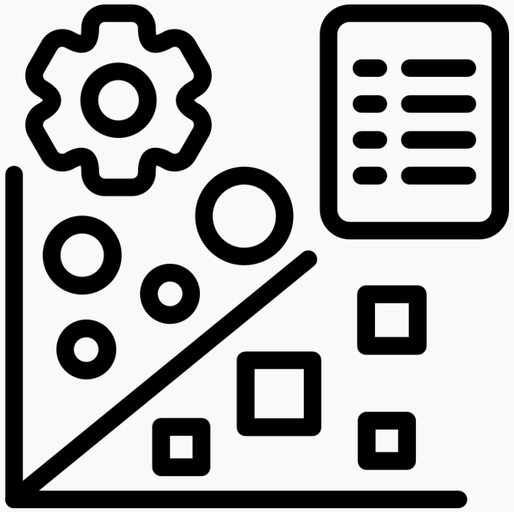
WHY SVM?

Provides a rigorous baseline for CRM data using maximum-margin separation.

The Kernel Trick: Maps non-linear sales engagement features into a separable high-dimensional space without heavy computation.

Convergence & Optimization: Guarantees global convergence via convex optimization, bounded exactly by the hardest-to-classify deals (Support Vectors).

Rigorous Assumptions: Relies appropriately on independent (i.i.d.) and stationary historical deal behaviors.



MODEL APPLICATION

DATA SETUP

- 8,000 samples
- Stratified split (Train / Val / Test)

BASELINE MODEL (TABULAR)

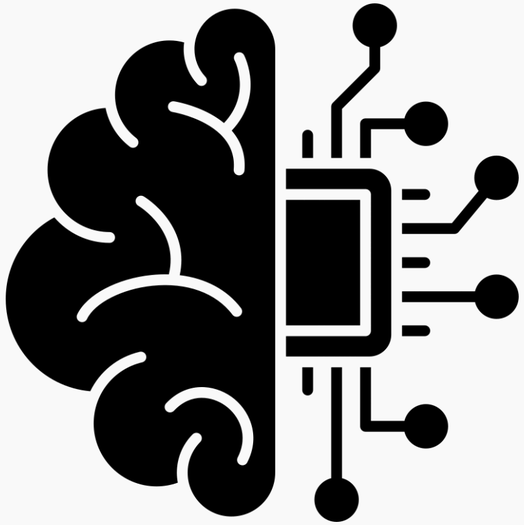
- SVM & Random Forest
- Accuracy: ~97%
- Key insight: Tabular features are highly predictive

DEEP LEARNING (TABULAR)

- MLP (64 → 32)
- Similar performance (~96.5%)
- No major advantage over baseline

EMBEDDINGS ONLY MODEL

- Text-only neural network
- Poor performance (~51% accuracy)
- Text alone is not sufficient



FINAL HYBRID MODEL

- Combined tabular + embeddings
- Best performance (97.6% accuracy)
- Improved stability & prediction quality

NOTEBOOK

RESULTS & EVALUATION

MODEL COMPARISON

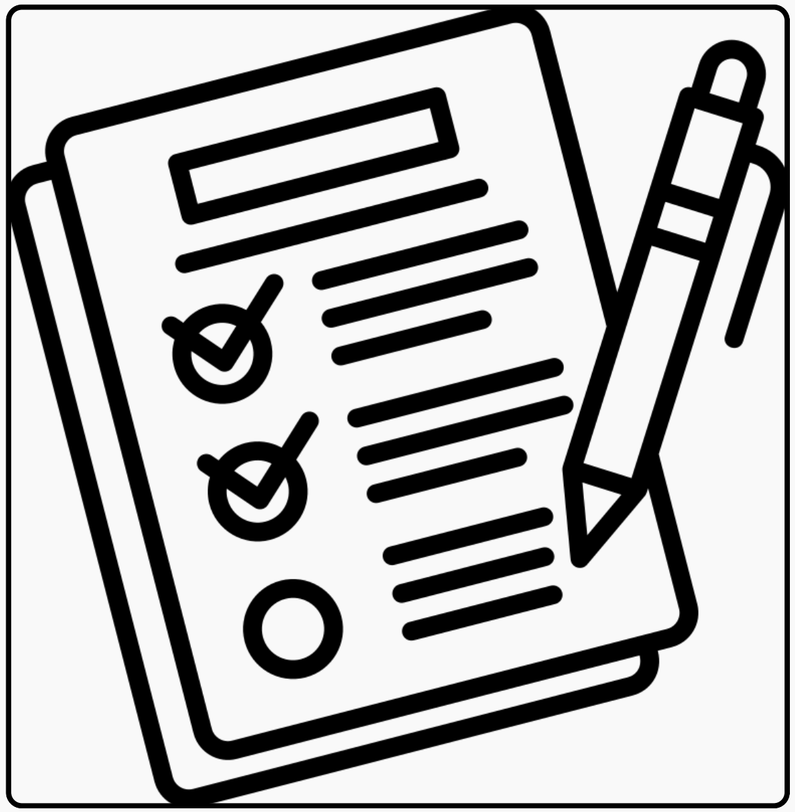
- Baseline (RF & SVM): ~97% Accuracy
- Tabular DL (MLP): 96.5% Accuracy
- Text-only Model: ~51% Accuracy (poor)
- Hybrid Model: 97.6% Accuracy (Best)

KEY OBSERVATIONS

- Tabular features alone are highly predictive
- Text embeddings alone are not sufficient
- Combining both → best performance

BEST MODEL PERFORMANCE

- Accuracy: 97.6%
- F1-Score: 0.976
- Precision: 0.976
- Recall: 0.977



KEY INSIGHTS

DATA INSIGHTS

- Tabular features are highly predictive
- Text alone is ambiguous and weak
- Balanced dataset → stable training

MODEL INSIGHTS

- Simple models (RF, MLP) perform very well
- Text-only models fail (~51% accuracy)
- Hybrid model → best performance

BUSINESS INSIGHTS

- Customer engagement strongly drives conversions
- Optimal engagement range: 0.6 – 0.75
- Conversation length has minimal impact

LIMITATIONS

- Text embeddings lack strong standalone predictive power
- Limited temporal modeling of conversations
- High-dimensional features increase model complexity

FUTURE WORK

- Use advanced NLP models (BERT, GPT-based)
- Incorporate sequential modeling (LSTMs / Transformers)
- Improve interpretability (SHAP, feature explanations)
- Build real-time prediction system

THANK YOU

[GET READY FOR PHASE-2]